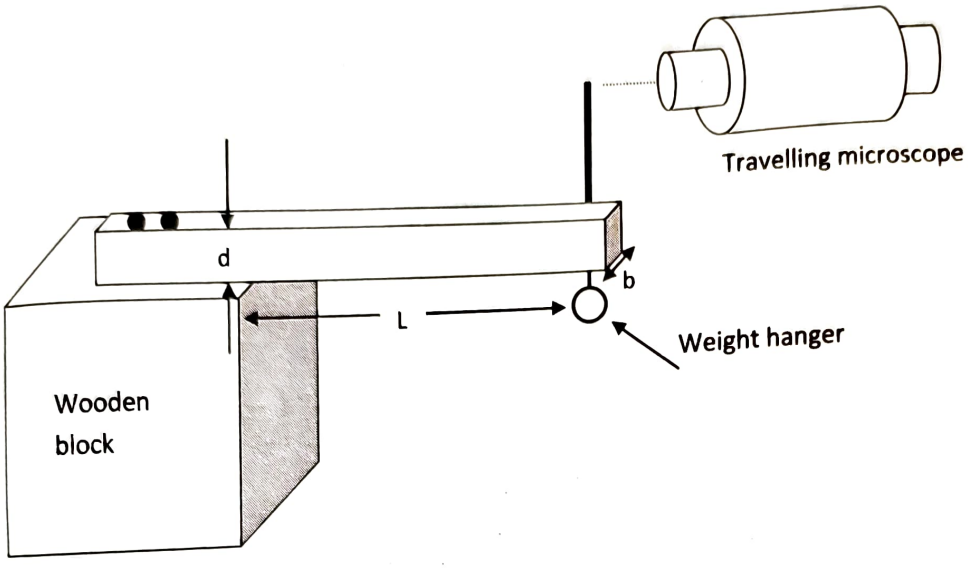
SINGLE CANTILEVER

OBSERVATIONS:

Experimental Setup:



Formula:

The Young's modulus of the material of the cantilever is calculated using the formula;

$$q = \frac{4mgL^3}{bd^3\delta_{mean}} \quad Nm^{-2}$$

Where, m = mass for which depression is measured

L is the length of the cantilever in metre

b is the breadth of the cantilever in metre

d is the thickness of the cantilevering metre

δ is the mean depression for =a mass m (40g) in metre

g is the acceleration due to gravity = 9.8 m/s

Least count of the travelling microscope

$$L.C = \frac{\text{Value of 1 main scale division}}{\text{Total number of vernier scale divisions}} = \frac{0.05}{50} = 0.001 \text{ cm}$$

SINGLE CANTILEVER

Experiment No:

Date: 21-11-2024

Aim: To determine the Young's modulus of the material of the single cantilever.

Apparatus: Thick rectangular metallic strip (cantilever), slotted weights with hanger, travelling microscope, screw gauge, Vernier callipers and metre scale etc.

Principle: Single cantilever is a beam which is fixed at one end and loaded at the other end. Young's modulus is one of the elastic constants which is a measure of elasticity of the material, it is defined as the ratio of longitudinal stress to the longitudinal strain within elastic limit. For a given cantilever, the depression produced at the loaded end of the cantilever depends on the load. The young's modulus (q) of the material of the cantilever is determined my measuring the depression produced for a given load at the free end of the cantilever.

Formula:

The Young's modulus of the material of the cantilever is calculated using the formula;

$$q = \frac{4mgL^3}{bd^3\delta_{mean}} \quad Nm^{-2}$$

Where, m = mass for which depression is measured

L is the length of the cantilever in metre

b is the breadth of the cantilever in metre

d is the thickness of the cantilevering metre

δ is the mean depression for a mass m(40g) in metre

g is the acceleration due to gravity = 9.8 m/s

Procedure:

- Suspend the weight hanger of mass w, at the free end of the cantilever. Adjust the vertical traverse of travelling microscope such that the cantilever and the microscope are in the same line
- Adjust the height of the microscope such that the pin tip in the cantilever is along the axis of the microscope.
- Make the object distance zero. Seeing through the eye piece gradually increase the object distance by moving the rack and pinion screw till a clear and inverted image of the pin tip is seen in the field of view.

Paste the data sheets here

Table 1: Travelling microscope readings for the load increasing.

Load (g)	MSR (cm)	CVD	TR= MSR + (CVD x LC) (cm)
W	6.3	10	6.3 + (10 × 0.001) = 6.3 cm
W + 10	6.2	2	6.2 + (2 × 0.001) = 6.202 cm
W + 20	6.1	11	6.1 + (11 × 0.001) = 6.111 cm
W + 30	6	28	6 + (28 × 0.001) = 6.028 cm
W + 40	5.9	37	5.9 + (37 × 0.001) = 5.937 cm
W + 50	5.85	4	5.85 + (4 × 0.001) = 5.854 cm
W + 60	5.75	23	5.75 + (23 × 0.001) = 5.773 cm
W + 70	5.70	38	5.70 + (38 × 0.001) = 5.738 cm

Table 2: Travelling microscope readings for the load decreasing.

Load (g)	MSR (cm)	CVD	TR= MSR + (CVD x LC) (cm)
W + 70	5.65	38	5.65 + (38 × 0.001) = 5.688 cm
W + 60	5.75	24	5.75 + (24 × 0.001) = 5.774 cm
W + 50	5.85	4	5.85 + (4 × 0.001) = 5.854 cm
W + 40	5.9	38	5.9 + (38 × 0.001) = 5.938 cm
W + 30	6	30	6 + (30 × 0.001) = 6.03 cm
W + 20	6.15	10	6.15 + (10 × 0.001) = 6.16 cm
W + 10	6.2	3	6.2 + (3 × 0.001) = 6.203 cm
W	6.3	10	6.3 + (10 × 0.001) = 6.31 cm

Table 3: Depression for a load of 40g (difference column)

Load (g)	TM reading (cm)		Mean TMR $R_L = (R_1 + R_2)/2$	Load (g)	TM readings (cm)		Mean TMR $R_R = (R_3 + R_4)/2$	Depression for 40g ($\delta = R_L \sim R_R$) cm
	Load increasing R_1	Load Decreasing R_2			Load Increasing R_3	Load Decreasing R_4		
W	6.31	6.31	6.31	W + 40	5.937	5.938	5.937	0.373
W + 10	6.202	6.203	6.202	W + 50	5.854	5.854	5.854	0.348
W + 20	6.111	6.118	6.135	W + 60	5.773	5.774	5.773	0.362
W + 30	6.028	6.03	6.029	W + 70	5.738	5.688	5.713	0.316

Mean depression for a load of 40 g = δ = 0.3495 cm.

- Adjust the height of the microscope such that the image of the pin tip touches with the horizontal cross wire and it is at the point of the intersection of cross wires.

- Note down the reading of the travelling microscope on the **vertical scale** and enter the reading in the table for load increasing (table-1) for the load w.

- Now add a mass of 10 g to the weight hanger and adjust the travelling microscope using the fine motion screw (vertical motion) to focus the tip of the pin with the point of intersection of cross wires. Note down the travelling microscope reading enter the reading in the table-1 for the load (w+10).

- Repeat the procedure by increasing the load in the weight hanger in steps of 10g up to a maximum load of w+70.

- Now repeat the experiment by decreasing the load in steps of 10g and note down the readings for w+70, w+60 up to w and enter the readings in the table for the load decreasing (table-2).

- Tabulate the readings of increasing and decreasing loads in table 3. Compute the mean value of the reading corresponding to each load and find the depression 'δ' for a load of 40 g (m) by a difference column method.

- Measure the length 'L' of the cantilever from the edge of the wooden block to the position of the pin using a metre scale.

- Determine the breadth 'b' of the cantilever using vernier callipers at different places on the metal strip and calculate the mean breadth.

- Determine the thickness 'd' of the cantilever using screw gauge at different places on the metal strip and find the mean thickness.

- Compute the young's modulus of the material of the cantilever by substituting the values of m, L, b, d and δ in the given formula.

- Calculate the fractional error $\Delta q/q$ and tabulate the result.

Note:

1. Level the travelling microscope using a spirit level.
2. Once you start the experiment do not shake or lean on the table, as this will disturb the focusing and affect the reading.
3. Add/ remove the weights gently on to/from the weight hanger
4. While performing the experiment care is to be taken to rotate the fine motion screw in only one direction so as to avoid backlash error.

Paste the data sheets here

Length of the cantilever, L = 17.6 cm.

Table 4: Breadth of the cantilever using vernier callipers

$$L.C = \frac{\text{Value of 1 main scale division}}{\text{Total number of vernier scale divisions}} = \frac{0.1}{10} = 0.01 \text{ cm}$$

Trial No.	MSR (cm)	CVD	TR = MSR + (CVD x LC) (cm)
1	2.3	8	2.3 + (8 × 0.01) cm
2	2.3	8	2.3 + (8 × 0.01) cm
3	2.3	8	2.3 + (8 × 0.01) cm
4	2.3	8	2.3 + (8 × 0.01) cm

Mean Breadth b = 2.38 cm

Table 5: Thickness (d) of the cantilever using screw gauge.

Least Count of Screw Gauge:

$$\text{Pitch} = \frac{\text{Distance moved on the pitch scale}}{\text{Total number of rotations given to screwhead}} = \frac{1}{1} = 1 \text{ mm}$$

$$L.C = \frac{\text{Pitch of the screw gauge}}{\text{Total number of head scale divisions}} = \frac{1}{100} = 0.01 \text{ mm}$$

Zero error (ZE) = $-20 \times 0.01 = -0.2$

Trial No.	PSR (mm)	HSD	TR= PSR + (HSD – (ZE)) LC in mm
1	0	63	0 + (63 − (−20)) × 0.01 = 0.83
2	0	63	0 + (63 − (−20)) × 0.01 = 0.83
3	0	63	0 + (63 − (−20)) × 0.01 = 083
4	0	63	0 + (63 − (−20)) × 0.01 = 083

Mean thickness, d = 0.83 mm

ERROR CALCULATION:

The Young's modulus of the material of the cantilever is calculated using the formula;

$$q = \frac{4mgL^3}{bd^3\delta_{mean}} \quad Nm^{-2}$$

Taking the logarithm, $\log q = \log (4mg) + 3 \log L - \log b - 3 \log d - \log \delta$

Taking Partial Derivatives and Using Modulus:

The relative error (or fractional error) in q is given by:

$$\frac{\Delta q}{q} = 3\frac{\Delta L}{L} + \frac{\Delta b}{b} + 3\frac{\Delta d}{d} + \frac{\Delta \delta}{\delta}$$

Substitute the least count values of length (L), breadth (b), thickness (d) and depression (δ) as ΔL, Δb, Δd, *and* $\Delta \delta$ respectively to calculate the value of $\Delta q/q$.

CALCULATIONS:

Mean TMR :

$w = \frac{6.31+6.31}{2} = 6.31$, $w+10 = \frac{6.202+6.203}{2} = 6.202$, $w+20 = \frac{6.114+6.116}{2} = 6.115$

$w+30 = \frac{6.028+6.03}{2} = 6.029$, $w+40 = \frac{5.937+5.938}{2} = 5.937$, $w+50 = \frac{5.854+5.854}{2} = 5.854$

$w+60 = \frac{5.773+5.774}{2} = 5.773$, $w+70 = \frac{5.738+5.688}{2} = 5.713$

Depression, δ

$\delta_1 = 6.31 - 5.937 = 0.373$ $\quad \delta_3 = 6.135 - 5.773 = 0.362$

$\delta_2 = 6.202 - 5.854 = 0.348$ $\quad \delta_4 = 6.029 - 5.713 = 0.316$

Mean Depression, $\delta = \frac{\delta_1+\delta_2+\delta_3+\delta_4}{4} = 0.3495 cm$

Mean Breadth, $b = \frac{2.38+2.38+2.38+2.38}{4} = 2.38 cm$

Mean thickness, $d = \frac{0.63+0.63+0.63+0.63}{4} = 0.63 mm$

$Y = \frac{4mgL^3}{bd^3\delta_{mean}} = \frac{4 \times 0.04 \times 9.8 \times (0.176)^3}{0.0238 \times (0.00063)^3 \times 0.0035} = 1.7973 \times 10^{11}$

26/30

Result:

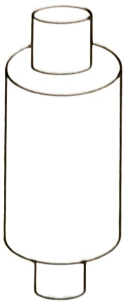
1. Young's modulus of the given material is found to be 1.7973×10^{11} N m^{-2}.
2. The relative error (or fractional error) in q is 3.629.

INTERFERENCE AT A WEDGE

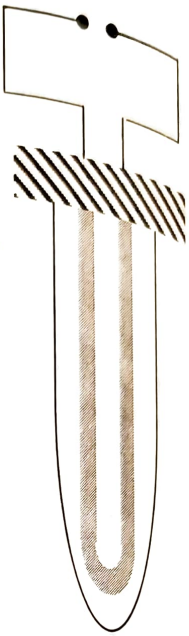
OBSERVATIONS



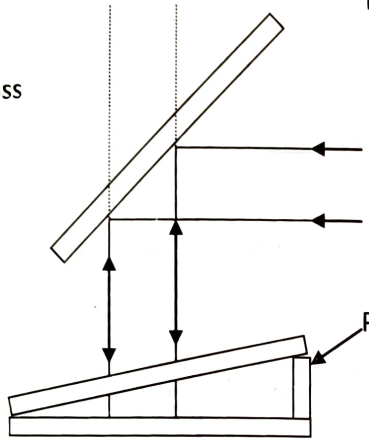
Fringe pattern system



Travelling microscope



Sodium vapor lamp



Inclined glass plate

Paper piece

Least count of the travelling microscope

$$LC = \frac{\text{Value of 1 main scale division}}{\text{Total number of vernier scale divisions}} = \frac{1}{100} \text{ cm}$$

Tabular column

Fringe number	TM readings, MSR + (CVD x LC)	Total reading (cm)
N	2·3 + (19 × 0·001)	2·319
n + 2	2·25 + (14 × 0·001)	2·264
n + 4	2·15 + (7 × 0·001)	2·157
n + 6	2·1 + (29 × 0·001)	2·129
n + 8	2·0 + (27 × 0·001)	2·027
n + 10	1·9 + (4 × 0·001)	1·904
n + 12	1·8 + (41 × 0·001)	1·841
n + 14	1·7 + (30 × 0·001)	1·73
n + 16	1·6 + (39 × 0·001)	1·639
n + 18	1·5 + (45 × 0·001)	1·545
n + 20	1·35 + (22 × 0·001)	1·372
n + 22	1·25 + (27 × 0·001)	1·277

INTERFERENCE AT A WEDGE

Experiment No. 16

Date: 28-11-2024

Aim: To determine the thickness of a thin paper piece by setting up interference fringes at an air wedge.

Apparatus: Two (optically flat) thick glass plates, sodium vapour lamp, travelling microscope (bridge type), a thin paper piece, reflecting glass plate, reading lens and a scale.

Principle:

A thin film with zero thickness at one end that gradually increases to a certain thickness at the other end is known as a wedge. An air wedge can be created by placing two glass slides together at one edge and separating them with a thin spacer (here paper sheet) at the opposite edge.

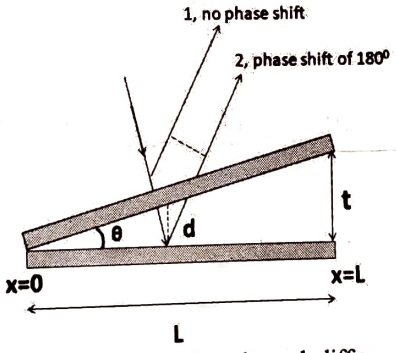
When a parallel beam of monochromatic light is directed nearly perpendicular to the wedge, it will be divided into two parts by the wave amplitude. One part reflects off the top surface of the air wedge (the lower surface of the upper glass slide), while the other part passes through to the bottom surface of the air wedge (the upper surface of the lower glass slide) and reflects there. Since the reflection is from an optically denser to a lower optical density medium, the reflected light waves will be phase shifted by 180^0 this is equivalent to an additional optical path difference of $\lambda/2$ between the two upper and lower reflected beams. The path difference, Δ, between the two reflected beams is given by:

$$\Delta = 2 d \mu \cos r - \lambda/2$$

Where:

- 'd' is the thickness of the air wedge at point of reflection.
- 'μ' is the refractive index of the medium between the plates (for air, $\mu \approx 1$)
- 'r' is the angle of refraction, for normal incidence $r = 0^0$

Therefore, $\Delta = 2d - \lambda/2$



Constructive interference (bright fringes) occurs when the path difference is an integer multiple of the wavelength λ of the light: $\Delta = 2d - \lambda/2 = n\lambda$ **(for n=0,1,2,...)**

Destructive interference (dark fringes) occurs when the path difference is an odd multiple of half the wavelength: $\Delta = 2d - \lambda/2 = (n+1/2)\lambda$ **(for n = 0,1,2,...)**

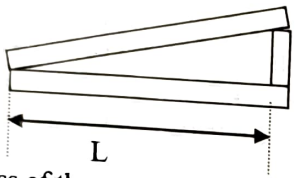
Since the thickness 'd' of the air wedge varies along its length, the path difference also changes, leading to constructive and destructive interference at different points. This results in a pattern of alternate equidistant, bright and dark fringes, which are parallel to the edge where the glass plates meet. The spacing between the fringes depends on the wavelength of the light and the angle of the wedge.

Table 2:

Fringe No.	TM Reading (cm), R_1	Fringe No.	TM Reading (cm), R_2	$X = 12\beta = R_2 - R_1$ (cm)
N	2.319	n + 12	1.841	0.478
n + 2	2.264	n + 14	1.73	0.534
n + 4	2.157	n + 16	1.639	0.518
n + 6	2.129	n + 18	1.545	0.584
n + 8	2.027	n + 20	1.372	0.655
n + 10	1.904	n + 22	1.277	0.627

The fringe width, β, is x/12 = 0.000471 m. Average x = 0.00566 m

To measure the length, L, of the air wedge



L = 0.064 m

The thickness of the paper is calculated by using the formula

$t = \frac{\lambda L}{2\beta}$ = 4.00×10^{-5} m

Where,

t is the thickness of the paper 4.00×10^{-5} m

L is the length of the air wedge 0.064 m

λ is the wavelength of sodium light = 589.3nm

β is the fringe width 0.000471m

Calculations

$t = \frac{\lambda L}{2\beta} = \frac{(589.3 \times 10^{-9}) \times 0.064}{2 \times 0.000471} = \frac{3.77 \times 10^{-8}}{9.42 \times 10^{-4}} = 4.00373 \times 10^{-5}$ m

Result:

The thickness of the given paper = 4.00373×10^{-5} m.

If white light is used instead of monochromatic light, the fringes appear coloured because different wavelengths interfere at different positions.

The fringe width (distance between consecutive fringes) 'β' is given by: $\beta = \lambda / 2\theta$

where: λ is the wavelength of the light, θ is the angle of the wedge (the angle between the two glass plates).

As θ is small it can be expressed as: $\theta = t / L$

where: 't' is the thickness of the paper or spacer, and 'L' is the length of the air wedge.

Thus, the fringe spacing β can be rewritten as: $\beta = \lambda L / 2t$

Therefore, the thickness of the paper (spacer) can be determined using: $t = \lambda L / 2\beta$

Procedure:

a) Experimental Setup

- The given optically plane glass plates and the reflecting glass plate are cleaned well by using muslin cloth before starting the experiment. The optically plane glass plates are kept one above the other. A thin rectangular paper piece is inserted between the glass plates at one extreme end, such that the piece of paper is perpendicular to the length of the glass plate. Now a wedge-shaped air film is formed in between the plates. The thickness of the air film gradually increases, from the point of contact of glass plate, till the inner edge of the paper piece.
- Now illuminate the air wedge by a parallel beam of sodium light using reflecting glass plate by making an angle of inclination 45° such that light after reflection will fall on the optical flats as shown in the diagram. The whole setup is kept below the objective of the travelling microscope such that the illuminated portion of the air wedge comes under the field of view of travelling microscope. Then focus the travelling microscope using rack and pinion screw such that the alternate dark and bright interference fringes are clearly seen.

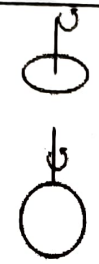
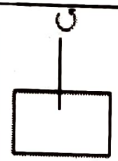
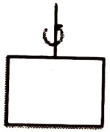
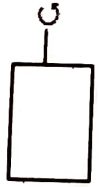
b) Measurement of fringe width

- The least count of travelling microscope is determined by finding the ratio of the value of one main scale division to the total number of vernier scale divisions.
- The vertical cross wire is adjusted to be at the center of any one of the dark fringes. The reading on the main scale and the coinciding vernier scale division are noted. Let this fringe be the nth dark fringe.
- The reading corresponding to the dark fringes n+2, n+4, n+6 up to n+22 are noted. The microscope may be moved either only to the left or the right side for this purpose.
- The readings are tabulated in the observation column and the fringe width is calculated.
- The length of the air wedge between the point of contact of the glass plates and the inner edge of the paper piece is measured using traveling microscope / scale.
- The thickness of the given paper is calculated using the formula. $t = \frac{\lambda L}{2\beta}$ m

$\frac{26}{30}$ for 5 M

Result: The thickness of the given paper = 4.00373×10^{-5} m.

TORSION PENDULUM – OBSERVATIONS

Body	M (Kg)	Dimension (m)	Axis	Moment of Inertia I (Kgm²)	Time for 10 osc. (s)	Mean time 't' for 10 oscillation. (s)	Period T=t/10 (s)	$\frac{I}{T^2}$ (Kgm²s⁻²)
Circular plate	1.49 Kg	Radius = 0.076m		$I_1 = \frac{MR^2}{2}$ = 4.3×10^{-3}	1. 38.13 2. 39.54 3. 35.54 4. 39.39	38.39	T_1 = 3.83	$\frac{I_1}{T_1^2}$ = 2.9×10^{-4}
				$I_2 = \frac{MR^2}{4}$ = 2.15×10^{-3}	1. 32.58 2. 33.30 3. 29.61	31.38	T_2 = 3.18	$\frac{I_2}{T_2^2}$ = 2.12×10^{-4}
Rectangular plate	1.247Kg	Length, L = 0.154m		$I_3 = \frac{M(L^2+B^2)}{12}$ = 3.53×10^{-3}	1. 35.97 2. 39.19 3. 38.34	37.83	T_3 = 3.78	$\frac{I_3}{T_3^2}$ = 2.47×10^{-4}
				$I_4 = \frac{ML^2}{12}$ = 3.795×10^{-4}	1. 26.44 2. 32.89 3. 31.91	30.41	T_4 = 3.041	$\frac{I_4}{T_4^2}$ = 4.10×10^{-4}
		Breadth, B = 0.105m		$I_5 = \frac{MB^2}{12}$ = 1.143×10^{-3}	1. 20.03 2. 19.69 3. 20.13	19.95	T_5 = 1.995	$\frac{I_5}{T_5^2}$ = 2.87×10^{-4}

$\left(\frac{I}{T^2}\right)_{mean}$ = 2.852×10^{-4} Kgm²s⁻²

TORSION PENDULUM

Experiment No: 7 Date: 5-12-2024

Aim: To determine the moment of inertia of the given irregular body and Rigidity modulus of the material of the give wire.

Apparatus and other materials required: Rectangular, circular and irregularly shaped plates, steel or brass wire, chuck nuts, stop clock, pointer, metre scale, Screw gauge weight box.

Principle:

Torsion pendulum is an angular harmonic oscillation. Moment of Inertia of a body is the reluctance to change its state of rest or uniform circular motion. A body whose moment of inertia I about an axis is known, is made to oscillate about the same axis, corresponding period T is noted. The ratio $\frac{I}{T^2}$ is a constant for different bodies and different axes **as long as the dimension of the suspension wire remains the same.** For a torsion pendulum $T=2\pi\sqrt{\frac{I}{C}}$ where C is the couple per unit twist of the wire and it is a constant. Hence, $I/T^2=C/4\pi^2$ is a constant and $C= 8\pi\eta r^4/2L$

Formula:

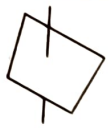
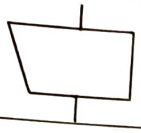
(a) Moment of inertia of irregular body about the axis through the CG and perpendicular to its plane, I'

$I_\alpha=\left(\frac{I}{T^2}\right)_{mean} \times T_\alpha^2 =$ 3.99 X 10^{-3} Kgm^2

(b) Moment of inertia of the irregular body about the axis through the CG and parallel to its plane

$I_\beta=\left(\frac{I}{T^2}\right)_{mean} \times T_\beta^2 =$ 3.08 X 10^{-3} Kgm^2

(c) Rigidity modulus $\eta=\left(\frac{8\pi L}{r^4}\right)\left(\frac{I}{T^2}\right)$ where r is the radius and L is the length of the wire.

Axis through CG	Time for 10 sec. (s)	Mean time(t) for 10 oscillations in s	Period T=t/10 s
	1. 34.15 2. 38.16 3. 39.24	37.183	T_α = 3.718
	1. 34.75 2. 31.07 3. 32.12	32.646	T_β = 3.264

Determination of Moment of Inertia of Irregular body

Formulae:

(a) Moment of inertia of irregular body about the axis through the CG and perpendicular to its plane, I

$$I_\alpha = \left(\frac{I}{T^2}\right)_{mean} \times T_\alpha^2 \quad Kgm^2$$

(b) Moment of inertia of the irregular body about the axis through the CG and parallel to its plane

$$I_\beta = \left(\frac{I}{T^2}\right)_{mean} \times T_\beta^2 \quad Kgm^2$$

Determination of rigidity modulus of the material of the wire

Radius of the wire (r) using screw gauge.

$$\text{Pitch} = \frac{\text{Distance moved on the pitch scale}}{\text{Total number of rotations given to screwhead}} = \frac{1}{1} = 1 \text{ mm}$$

$$\text{L.C} = \frac{\text{Pitch of the screw gauge}}{\text{Total number of head scale divisions}} = \frac{1}{100} = 0.01 \text{ mm}$$

Zero error (ZE) = +2.

Paste the data sheets here

Procedure:

- Measure the dimensions of the given circular and rectangular discs.
- Clamp one end of the wire through the chuck nut to a regular disc and other end to the top end of the retard stand.
- Twist the wire through a small angle and then let free so that the body executes torsional oscillations **(The oscillations should be in a horizontal plane. Arrest the side ward movement or wobbling if any)**.
- For each configuration of the pendulum, note down the time taken for 10 oscillations and repeat the process thrice. Tabulate this in table 1.
- Calculate mean time (t) for 10 oscillations and hence the time period T then find $\frac{I}{T^2}$ for each axis. Follow the same procedure for the axes of the other regular body
- Repeat the procedure for two mutually perpendicular axes of the irregular body, determine the average period of oscillation for two different axes and tabulate values in table 2.
- Find out the moment of inertia of irregular body using given formulae.
- Measure the diameter of the wire using screw gauge and enter the readings in the tabular column.
- Calculate the average diameter and radius of the wire.
- Measure the length L of the wire between the check nuts.
- Calculate the rigidity modulus of the material of the wire using the given formula.

Paste the data sheets here

Trial No.	PSR(mm)	HSD	TR= PSR + (HSD – (ZE))LC (mm)
1	0	78	$0 + (78-2) \times 0.01 = 0.76$
2	0	80	$0 + (80-2) \times 0.01 = 0.78$
3	0	79	$0 + (79-2) \times 0.01 = 0.77$
4	0	80	$0 + (80-2) \times 0.01 = 0.78$

Mean diameter of the wire, d = 0.7725 mm 7.725×10^{-4} m

Mean Radius of the wire, r = 3.8625×10^{-4} m

Length of the wire, L = 0.238 m

Rigidity modulus $\eta = \left(\frac{8\pi L}{r^4}\right)\left(\frac{I}{T^2}\right)$

$= \left(\frac{8 \times 3.14 \times 0.238}{(3.86 \times 10^{-4})^4}\right)(2.852 \times 10^{-4})$

$= \left(\frac{5.978}{2.228 \times 10^{-14}}\right) \times (2.829 \times 10^{-4})$

$= 7.77005 \times 10^{10}$

Result:

1. The moment of inertia of irregular body about an axis perpendicular to the plane $I_\alpha = 3.99 \times 10^{-3}$ Kgm^2.
2. The moment of inertia of irregular body about an axis parallel to the plane $I_\beta = 3.08 \times 10^{-3}$ Kgm^2.
3. Rigidity modulus of the material of the wire 'η' = 7.77×10^{10} N/m^2

CALCULATIONS:

$I_1 = \frac{MR^2}{2} = \frac{(1.491)\times(0.0761)^2}{2} = 4.3\times10^{-3}$

$\frac{I_1}{T_1} = \frac{4.3\times10^{-3}}{(3.83)^2} = 2.9\times10^{-4}$

$I_2 = \frac{MR^2}{4} = \frac{(1.491\times(0.0761)^2}{4} = 2.15\times10^{-3}$

$\frac{I_2}{T_2} = \frac{2.15\times10^{-3}}{(3.181)^2} = 2.12\times10^{-4}$

$I_3 = \frac{M(L^2+B^2)}{12} = \frac{1.247[(0.154)^2+(0.105)^2]}{12} = 3.53\times10^{-3}$

$\frac{I_3}{T_3^2} = \frac{3.53\times10^{-3}}{(3.78)^2} = 2.47\times10^{-4}$

$I_4 = \frac{ML^3}{12} = \frac{1.247\times(0.154)^3}{12} = 3.795\times10^{-4}$

$\frac{I_4}{T_4^2} = \frac{3.795\times10^{-4}}{(3.041)^2} = 4.10\times10^{-4}$

$I_5 = \frac{MB^2}{12} = \frac{1.247\times(0.105)^3}{12} = 1.43\times10^{-3}$

$\frac{I_5}{T_5^2} = \frac{1.43\times10^{-3}}{(1.955)^2} = 2.87\times10^{-4}$

$\left(\frac{I}{T^2}\right)_{mean} = \frac{[(2.9\times10^{-4})+(2.12\times10^{-4})+(2.4\times10^{-4})+(4.10\times10^{-4})+(2.87\times10^{-4})]}{5}$

$= 2.892\times10^{-4}\ Kgm^2s$

$I_\alpha = \left(\frac{I}{T^2}\right)_{mean}\times T_\alpha^2\ Kgm^2 = 2.892\times10^{-4}\times(3.718)^2 = 3.99\times10^{-3}$

$I_\beta = \left(\frac{I}{T^2}\right)_{mean}\times T_\beta^2\ Kgm^2 = 2.892\times10^{-4}\times(3.214)^2 = 3.08\times10^{-3}$

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Result:

1. The moment of inertia of irregular body about an axis perpendicular to the plane $I_\alpha = 3.99\times10^{-3}$ Kgm^2.

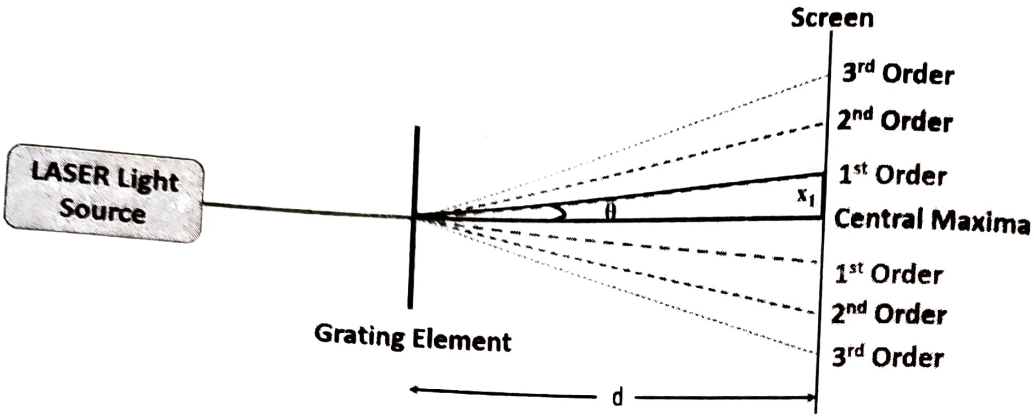
2. The moment of inertia of irregular body about an axis parallel to the plane $I_\beta = 3.08\times10^{-3}$ Kgm^2.

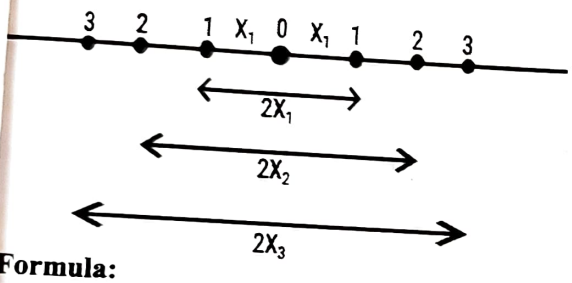
1. Rigidity modulus of the material of the wire 'η' = 7.77×10^{10} N/m^2

LASER DIFFRACTION

OBSERVATIONS:

Diagram:





Diffraction angle θ_n, $\tan\theta_n = \frac{x_n}{d}$

X_n = Distance between the central maximum and the nth order secondary maximum.

d = Distance between grating and the screen

Formula:

Wavelength of Laser source $\lambda = \frac{C \sin\theta_n}{n}$ ..... m

Where C is the grating constant, n is the order of the maximum, θ is the angle of diffraction

Grating Constant: $C = \frac{1\,\text{inch}}{\text{No. of lines (N) per inch}} = \frac{2.54\times10^{-2} m}{500} = 5.08\times10^{-5}\,\text{m}$

Distance between the grating and the screen, d = 7 cm

LASER DIFFRACTION

Experiment No: 7 Date: 7-11-2024

Aim: To determine the wavelength of a given laser beam

Apparatus: Laser source, Grating, Optical bench with accessories and metre scale etc.,

Principle: The principle of the experiment is diffraction and it is the bending of a waves round the corners of an obstacle, its effects are well observed, if the wavelength is comparable with the size of the obstacle. Grating is a plane glass plate on which equidistant, parallel lines are ruled with a diamond tip. The ruled portion is opaque and the space between the rulings acts as slit. The number of lines are ruled such that the width of the line is of the order of wavelength of the light used. When the laser beam is incident on the grating the rulings on the grating will act as obstacles and the laser light will undergo diffraction. The diffraction pattern consists of a central maximum and secondary maxima and minima on either side of the central maximum. On a screen placed behind the grating the diffraction pattern is seen with a central bright laser spot (the central maximum) and bright spots of decreasing brightness (secondary maxima) on either side of the central bright spot. The angle of diffraction is measured and it is used to calculate the wavelength of the laser light.

Formula:

Wavelength of laser light, $\lambda = \frac{C \sin\theta_n}{n}$ ..... m

Where C is the grating constant and it is the distance between corresponding points of two successive lines on the grating, n is the order of the maximum, θ_n is the angle of diffraction of the nth maximum (in degrees).

Procedure:

- Mount the laser on an upright stand at one end of the optical bench. Attach a screen to another upright stand and place it at the opposite end of the bench.
- Mark four quadrants on a graph sheet and mark the origin O, fix the graph sheet to the screen. Adjust the graph sheet's position so that the centre of the laser spot aligns with the origin 'O'.
- Mount the grating holder on an upright stand between the laser source and the screen.
- Mount the grating in the grating holder such that the lines on the grating are vertical (i.e the length of the grating is horizontal on the grating stand).

Table:

Diffraction order (n)	Distance $2X_n$(cm)	Distance X_n(cm)	Diffraction angle (θ_n in degree) $\theta_n = \tan^{-1}\left(\frac{X_n}{d}\right)$	Wavelength λ (nm) $\lambda = \frac{C \sin\theta_n}{n}$	Fractional error $\left(\frac{\delta\lambda}{\lambda}\right)$
1.	2 (1.8)	1 (0.9)	0.6829	6.05×10^{-5}	0.11
2.	4 (3.6)	2 (1.8)	1.3657	6.0537×10^{-5}	0.05
3.	6 (5.4)	3 (2.7)	2.0481	6.0517×10^{-5}	0.03
4.	8 (7.6)	4 (3.8)	2.881	6.0383×10^{-5}	0.02
5.	10 (9.8)	5 (4.9)	3.7133	6.580×10^{-5}	0.02
Average wavelength of the laser and fractional error				$\lambda = 6.226$ nm	$\left(\frac{\delta\lambda}{\lambda}\right) = 0.046$

Error Formula:

The formula used wavelength of laser light is $\lambda = \frac{C \sin\theta_n}{n}$

Taking the logarithm, $\log \lambda = \log (C/n) + \log(\sin\theta)$

Simplifying, we get the relative uncertainty (or fractional uncertainty) in λ is given by:

$$\frac{\delta\lambda}{\lambda} = \frac{\delta x_n}{x_n} + \frac{x_n \delta x_n + d\delta d}{\left(x_n^2 + d^2\right)} \approx \frac{\delta x_n}{x_n}$$

Substitute the least count values of (x_n) distance between the central maxima and the order of diffraction and (d) distance between grating and the screen as δx_n and δd, respectively to calculate the value of $\delta\lambda / \lambda$.

Calculations:

For λ,

$\lambda = \frac{C \sin\theta_n}{n}$ where C = Grating constant $= 5.08 \times 10^{-5}$ m

n=1, $\lambda_1 = \frac{5.08 \times 10^{-5} \times 0.6829}{1} = 6.05 \times 10^{-5}$

n=2, $\lambda_2 = \frac{5.08 \times 10^{-5} \times 1.3657}{2} = 6.0537 \times 10^{-5}$

n=3, $\lambda_3 = \frac{5.08 \times 10^{-5} \times 2.0481}{3} = 6.0517 \times 10^{-5}$

n=4, $\lambda_4 = \frac{5.08 \times 10^{-5} \times 2.881}{4} = 6.383 \times 10^{-5}$

n=5, $\lambda_5 = \frac{5.08 \times 10^{-5} \times 3.7133}{5} = 6.580 \times 10^{-5}$

Average $= \frac{\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5}{5}$

$\lambda = 6.226$ nm

Result: 1. The wavelength of laser light is found to be.....6.226.....nm

2. The relative error (or fractional error) in λ is.....0.046.....

- Fix the grating stand at some distance d from the screen and adjust the plane of the grating perpendicular to the incident laser light (i.e normal incidence). If the laser spots are on the horizontal line of the graph sheet the incidence is normal. Note down the distance '*d*' between the grating and the screen.
- Mark the centres of the central maximum and all the secondary maxima seen on the graph sheet using a pencil and remove the graph sheet from the stand.
- Measure the distance between the first-order secondary maxima on either side of the central maximum as $2X_1$. For the second-order secondary maxima, measure the distance as $2X_2$, and repeat this process for all pairs of maxima on the screen.
- By using the grating constant C and the angle of diffraction θ_n, calculate the wavelength of laser light for all the orders. Finally, find the average value of wavelength.
- Calculate the fractional error $\delta\lambda/\lambda$ and tabulate the result.

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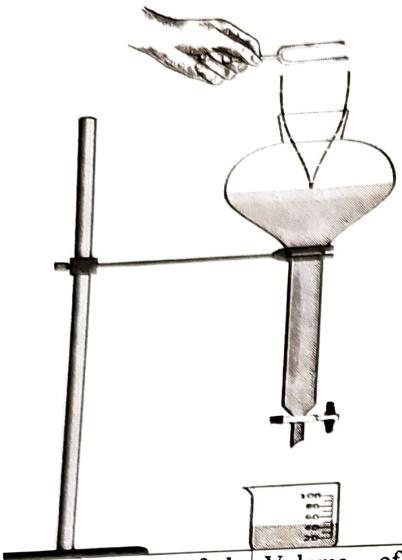
Result: 1. The wavelength of laser light is found to be.....6.226.....nm

2. The relative error (or fractional error) in λ is.....0.046....

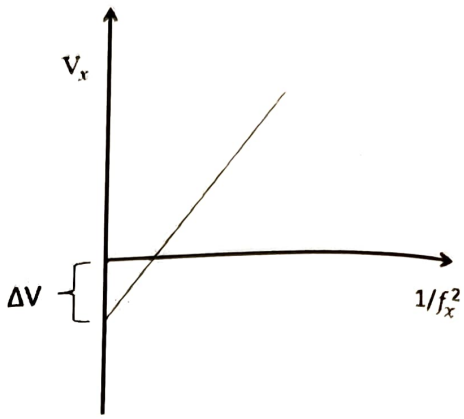
VOLUME RESONATOR

OBSERVATIONS

Experimental setup:



Model Graph



Sl.No.	Tuning fork frequency f in Hz	Volume of resonating air (water collected) in m^3			$1/f^2$	$f^2.(V+\Delta V)$ in $m^3\ s^{-2}$	$[f^2(V+\Delta V)]_{avg}$
		V_1 in m^3	V_2 in m^3	Mean volume V in m^3			
1							
2							
3							
4	$f_{unknown}$			V_f		$(V_f+\Delta V)=$	

Determining unknown frequency of the tuning fork

$$[f^2(V+\Delta V)]_{avg} = f^2_{unknown}(V_f+\Delta V)_{unknown}$$

$$f_{unknown} = \sqrt{\frac{[f^2(V+\Delta V)_{avg}]}{V_f+\Delta V}}\ Hz$$

Where, $f_{unknown}$ is the unknown frequency of the tuning fork, f = Known frequency of the tuning fork, V = volume of air resonating with the fork of known frequency, V_f = volume of air resonating with the fork of unknown frequency and ΔV is the volume correction obtained from graph.

VOLUME RESONATOR

Experiment No:

Date:

Aim: To determine the frequency of a tuning fork using volume resonator.

Apparatus: Aspirator bottle, tuning forks, rubber pad, beaker, measuring jar.

Principle: When the sound wave propagates through a medium the particles in the medium vibrate, causing variations in density and pressure along the wave's direction of travel. These changes result in a series of high-pressure and low-pressure regions. If the source of the sound waves vibrates sinusoidally, the pressure variations are also sinusoidal. When two waves of equal wavelength and amplitude propagating in opposite directions superimpose on each other, then interference occurs and the resultant wave is called a standing wave. In a standing wave the particles of the medium at certain points do not oscillate. These points are called nodes. At certain points the particles of the medium have maximum amplitude of oscillation, called antinodes. When the frequency of oscillation of the vibrating object is equal to the natural frequency of the system, then resonance occurs.

When the vibrating tuning fork is kept over the open end of a partially filled resonator, then the sound waves propagating in the resonator are reflected back from the surface of the water. A standing wave pattern is generated. If the frequency of the tuning fork is equal to the natural frequency of the air column inside the resonator, then the node of the standing wave pattern is produced at the water surface inside the resonator and the antinode is produced near the open end of the resonator.

The resonant frequency of a Helmholtz resonator can be derived by analyzing the oscillatory behavior of the air inside the resonator. This behavior is similar to that of a mass-spring system where the air in the neck acts like a mass and the air in the cavity acts like a spring.

The expression for the resonant frequency is given by:

$$f = \frac{c}{2\pi}\sqrt{\frac{A}{L_{eff}V_{eff}}} \quad \text{Hz}$$

Where f is resonant frequency, c is the speed of sound, A is the cross-sectional area of the neck, and the L_{eff} and V_{eff} are effective length and volume of the cavity and neck respectively.

$L_{eff} = L + 0.6\,R$, R is the radius of the neck of the bottle

$V_{eff} = V + \Delta V$, ΔV is the volume correction, it accounts for the additional effective volume and is added to the geometric volume V of the cavity.

$$f^2 = \frac{c^2}{4\pi^2}\frac{A}{L_{eff}V_{eff}} \qquad f^2 = \frac{Ac^2}{4\pi^2 L_{eff}}\frac{1}{V_{eff}} = Constant.\frac{1}{V_{eff}}$$

$$f^2 V_{eff} = Constant \qquad f^2 V_{eff} = f^2(V + \Delta V) = constant$$

$$V = \frac{Constant}{f^2} - \Delta V$$

The intercept of the plot V v/s $1/f^2$ gives 'ΔV' the volume correction value.

Need for volume correction:

When the Helmholtz resonator is excited, the air in the cavity is compressed and rarefied as the air in the neck moves back and forth. The volume of air inside the cavity that actually contributes to the oscillatory motion is not limited to the geometric volume of the cavity. Some of the air just outside the cavity near the neck also gets involved in the oscillation due to the pressure fluctuations. This means that the effective volume of the resonator is slightly larger than the simple geometric volume.

Procedure: The aspirator bottle is filled with water. The tuning fork of known frequency is excited and it is held above the mouth of the aspirator bottle as shown in the figure. At the same time, water in the bottle is allowed to run out of aspirator bottle by opening the pinch cock attached to the out let tube. Water is allowed to flow out till the air in the resonator is in resonance with the tuning fork used. In the correct position of resonance, a note of maximum intensity will be heard. At the same moment the flow of water should be stopped using the pinch cock. The volume 'V' of water is collected in a beaker and measured using measuring jar. The volume is equal to the volume of air in the resonator, which is in resonance with the tuning fork, when vibrated. The experiment is repeated for the same tuning fork and the mean value of volume 'V' is found. The experiment is repeated for different tuning fork as above and volume is noted. The volume of air 'V_f' is found out with the fork of unknown frequency $f_{unknown}$.

Plot a graph of volume 'V' along y-axis and $1/f^2$ along x-axis and find out the value of y-intercept 'ΔV' from the resulting graph.

Formula:

$$f_{unknown} = \sqrt{\frac{[f^2\,(V+\Delta V)_{avg}]}{V_f+\Delta V}}\ Hz$$

Where, $f_{unknown}$ is the unknown frequency of the tuning fork, f = Known frequency of the tuning fork, V = volume of air resonating with the fork of known frequency, V_f = volume of air resonating with the fork of unknown frequency and ΔV is the volume correction obtained from graph.

Result:

Volume correction = ΔV =

The unknown frequency of given tuning fork =_____Hz.

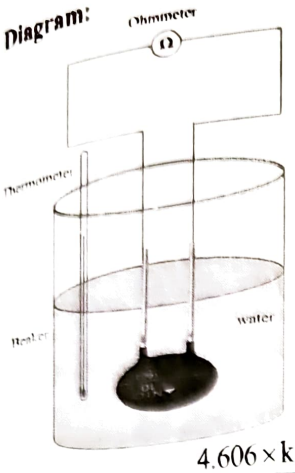
Note: Start the experiment by choosing higher frequency tuning fork and proceed the experiment in decreasing order of their frequencies.



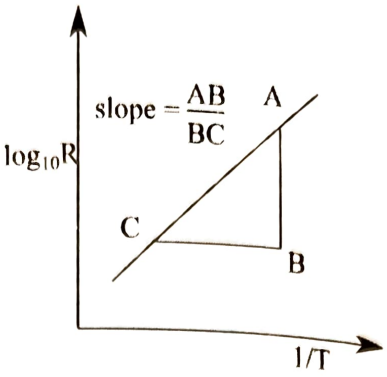
BAND GAP OF A THERMISTOR

OBSERVATIONS:

Diagram:



Model Graph:



Formula: $E_g = \dfrac{4.606 \times k \times m}{1.6 \times 10^{-19}}$ eV

Where E_g = Energy gap of a given thermistor in eV

k = Boltzmann constant = 1.381×10^{-23} J/K

m = Slope of the graph

Table:

Sl. No	Temp t °C	Temp T(K)	R (Ω)	log R	1/T ($\times 10^{-3}$ K^{-1})
Room Temp	27.8	300.95	35.8	1.55	3.3228 $\times 10^{-3}$
1.	85.0	358.15	6.7	0.8210	2.7921 $\times 10^{-3}$
2.	83.1	356.25	6.8	0.8325	2.8070 $\times 10^{-3}$
3.	81.1	354.25	7.1	0.8512	2.8228 $\times 10^{-3}$
4.	79.1	352.25	7.5	0.8750	2.8388 $\times 10^{-3}$
5.	77.9	351.05	7.7	0.8864	2.8485 $\times 10^{-3}$
6.	76.1	349.25	8.1	0.9084	2.8632 $\times 10^{-3}$
7.	74.0	347.15	8.5	0.9294	2.8805 $\times 10^{-3}$
8.	72.1	345.25	8.8	0.9444	2.8964 $\times 10^{-3}$
9.	70.3	343.45	9.2	0.9637	2.9116 $\times 10^{-3}$
10.	68.2	341.35	9.8	0.9912	2.9295 $\times 10^{-3}$

CALCULATIONS:

$E_g = \dfrac{4.606 \times k \times m}{1.6 \times 10^{-19}}$ eV $= 0.462237$

$k = 1.381 \times 10^{-23}$

$m = 1.1627 \times 10^{3}$

Result: The energy gap (band gap) of the given thermistor is 0.462237 eV.

BAND GAP OF A THERMISTOR

Experiment No: Date: 14-11-2024

Aim: To determine the energy gap (E_g) of a Thermistor.

Apparatus: Glass beaker, Thermistor, Multi meter, Thermometer.

Principle: A thermistor is a thermally sensitive resistor. Thermistors are made of semiconducting materials such as oxides of Nickel, Cobalt, Manganese and Zinc. They are available in the form of beads, rods and discs.

The variation of resistance of thermistor is given by $R = ae^{\frac{b}{T}}$ where 'a' and 'b' are constants for a given thermistor, b is a measure of the band gap. The resistance of thermistor decreases exponentially with rise in temperature. At absolute zero all the electrons in the thermistor are in valence band and conduction band is empty. As the temperature increases electrons jump to conduction band and the conductivity increases and hence resistance decreases. By measuring the resistance of thermistor at different temperatures the energy gap is determined.

Formula: $E_g = \frac{4.606 \times k \times m}{1.6 \times 10^{-19}} eV$

Where, E_g = Energy gap of a given thermistor in eV, k = Boltzmann constant = 1.381 x 10^{-23} J/K .

m = Slope of the graph of log R vs 1/T.

Procedure:

- Make the circuit connection as shown in the figure.
- Keep the multi meter in resistance mode (200 Ω range).
- Insert the thermometer in a beaker containing tap water, thermistor and note down the resistance at room temperature.
- Immerse the thermistor in hot water at about 90°C.
- Note down the resistance of the thermistor for every decrement of 1°C in the beginning and a decrement of 2^0 up to 60°C.
- Plot the graph of log R versus 1/T and calculate the slope 'm'.
- Calculate the energy gap of a given thermistor using relevant formula.

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Result: The energy gap (band gap) of the given thermistor is 0.482237 eV.